Feature Selection & Agro-Climatic Engineering Report

Project: AI-Powered Crop Pest Prediction | Date: 2026-10-05 | Senior Data Scientist

1. Feature Selection Methodology

Features were selected based on biological realism, pre-prediction availability, mutual information gain, and tree feature importance. 15 raw inputs and 4 scientifically sound engineered features constitute the final 19-dimensional input space.

2. Selected Final Feature Taxonomy

Feature Name	Type	Domain Source	Agronomic Significance
District	Categorical	Farmer / Location	Geographic microclimate zone
Crop_Name	Categorical	Farmer Input	Primary host plant specificity
Season	Categorical	Seasonal Calendar	Agricultural cultivation cycle (Kharif/Rabi)
Soil_Type	Categorical	Soil Survey / Farmer	Soil water retention and drainage capacity
Temperature	Numerical (°C)	Weather API / Sensor	Controls insect metabolic rate and degree-days
Humidity	Numerical (%)	Weather API / Sensor	Governs egg hatching and fungal spore viability
Rainfall	Numerical (mm)	Weather API / Sensor	Determines vegetative growth & larval survival
Wind_Speed	Numerical (km/h)	Weather API / Sensor	Influences aerial pest dispersal and migration
Soil_Moisture	Numerical (%)	Soil Sensor / Defaults	Regulates pupation survival in soil
Soil_pH	Numerical	Soil Health Card	Affects nutrient uptake and host resilience
NPK (N, P, K)	Numerical (kg/ha)	Soil Health Card	Primary soil macronutrient concentrations
Organic_Carbon	Numerical (%)	Soil Health Card	Soil organic matter and biological health
Temp_Humidity_Index	Engineered	Thom's THI Formula	Bio-meteorological heat stress & pest comfort
Weather_Score	Engineered	Microclimate Gaussian	Composite multi-parameter suitability score
NPK_Total	Engineered	N + P + K	Net chemical fertility burden
NPK_Ratio	Engineered	(N+P) / (K+1)	Vegetative softness vs cell wall defense ratio